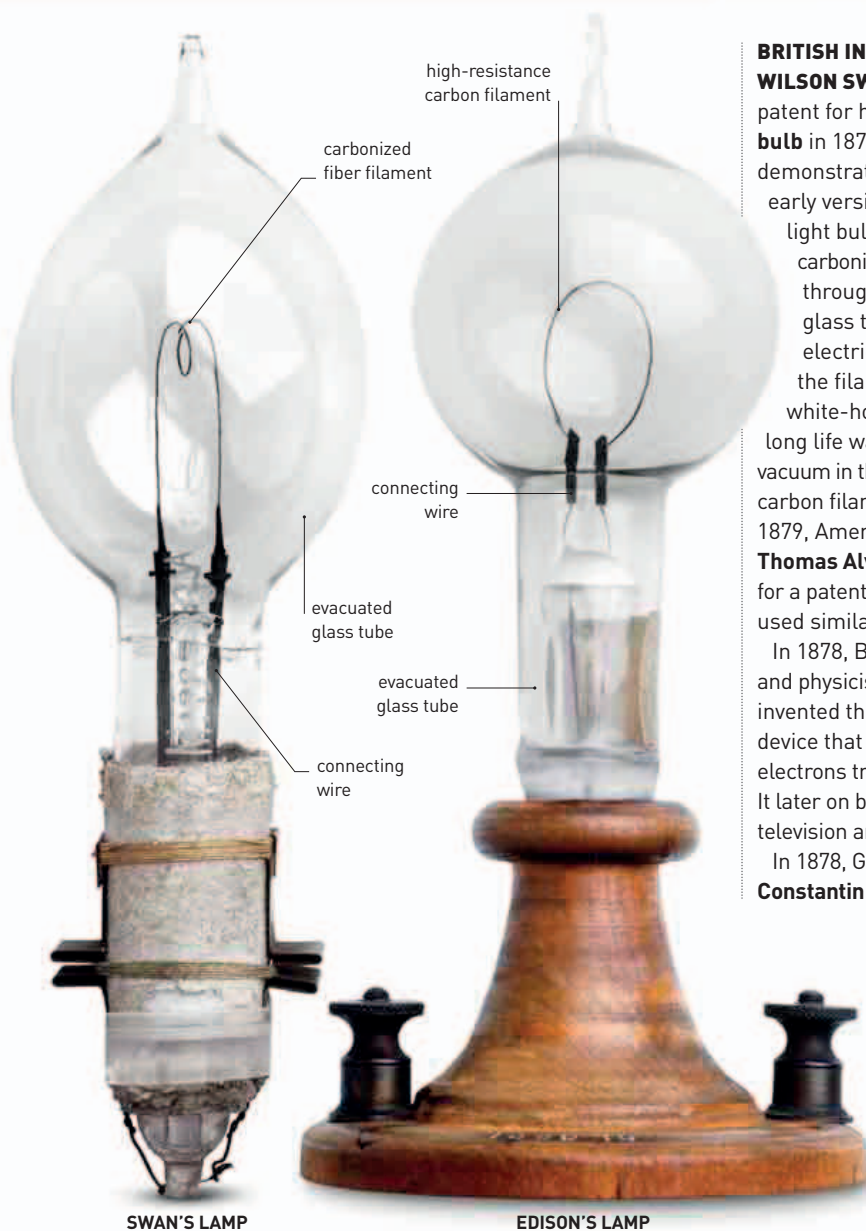


8 MILES PER HOUR THE SPEED OF THE SIEMENS LOCOMOTIVE

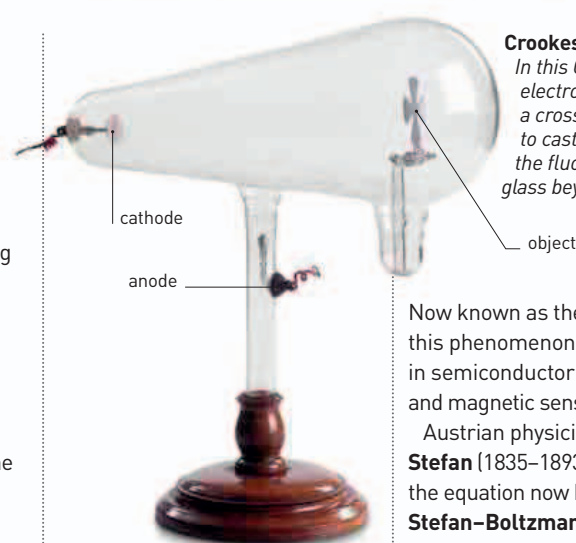


This photograph shows people traveling on carriages pulled by the Werner–Siemens locomotive, which was demonstrated at the Berlin Trade Fair in 1879. This locomotive was the first to be powered by electricity.



BRITISH INVENTOR JOSEPH WILSON SWAN received a patent for his **electric light bulb** in 1878, which he had demonstrated in 1860. This early version of the modern light bulb consisted of a carbonized filament running through an evacuated glass tube. Passing an electric current through the filament made it glow white-hot. The keys to its long life was an improved vacuum in the glass tube and the carbon filament. In November 1879, American inventor **Thomas Alva Edison** applied for a patent for a light bulb that used similar technology. In 1878, British chemist and physicist **William Crookes** invented the **Crookes tube**, a device that helped show that electrons travel in straight lines. It later on became the basis of television and other displays. In 1878, German chemist **Constantin Fahlberg** (1850–1910)

Early light bulbs
The lamps developed by Swan and Edison were almost identical. After a legal battle over rights, the two inventors formed the Edison–Swan company to jointly market the bulbs.



Crookes tube
In this Crookes tube, electrons stream past a cross-shaped object to cast a shadow on the fluorescent glass beyond.

Now known as the **Hall effect**, this phenomenon is important in semiconductor technology and magnetic sensors.

Austrian physicist **Josef Stefan** (1835–1893) formulated the equation now known as the **Stefan–Boltzmann law**, which concerned the calculation of radiation emitted from black bodies—surfaces that absorb all the electromagnetic radiation that strikes them. In 1884, Stefan's colleague Ludwig Boltzmann (1844–1906) explained the law using thermodynamics.

American scientist **Albert Abraham Michelson** (1852–1931) measured the **speed of light**

accidentally discovered a natural sweetener, later called **saccharin**, while working with coal tar in the US. Saccharin is about 200 times sweeter than sugar.

German engineer **Karl Benz** (1844–1929) developed a one-cylinder **two-stroke gas engine**, demonstrated for the first time on December 31, 1879.

In his 1879 book, *Cartography of Russian Soils*, Russian geologist **Vasily Dokuchaev** (1846–1905) introduced the concept of **pedology**, the scientific study of soil.

American physicist **Edwin Hall** (1855–1938) discovered that a magnetic field created at a right angle to the flow of electric current would create a voltage difference across the conductor.

“ WE WILL MAKE ELECTRICITY SO CHEAP THAT ONLY THE RICH WILL BURN CANDLES. ”

Thomas Edison, American inventor, 1879

1878 Joseph Wilson Swan receives a patent for the electric light bulb

1878 Gustav Kessel patents an early espresso machine

1878 British engineer Dugald Clerk designs a forerunner of the two-stroke diesel engine

1878 American scientist Josiah Willard Gibbs publishes pioneering research on thermodynamics and entropy

February 27, 1879 Constantin Fahlberg accidentally discovers the artificial sweetener saccharin

1879 Edwin Hall discovers the Hall effect, a phenomenon used in semiconductor technology and magnetic sensors

1879 Albert Abraham Michelson measures the speed of light

1878 Emil Hermann Fischer works out the chemical formula for phenylhydrazine

1878 Lord Rayleigh publishes the first volume of *The Theory of Sound*

1879 Karl Benz receives a patent for the two-stroke gas engine

1879 Vasily Dokuchaev publishes *Cartography of Russian soil*

1879 Josef Stefan develops what came to be known as the Stefan–Boltzmann law

1879 Werner von Siemens builds the first electric locomotive